

q+a link

<https://tinyurl.com/140zoom>



Lecture 2: Working memory

PSYCH/COGS 140M 2026.04.07

What is Working Memory?

Outline

- What is working memory (WM)?
- WM capacity
- Separate WM and LTM systems?

Outline

- What is working memory (WM)?
- WM capacity
- Separate WM and LTM systems?

Multiple memory systems

Transient memory

Sensory Short-term
(working)

Long-term memory

Declarative

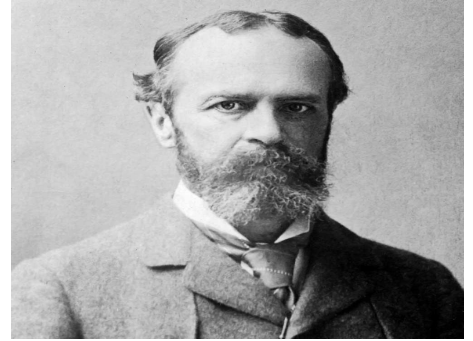
Semantic Episodic

Non-declarative

Conditioning Skill-learning Priming

William James

- Proposed two forms of memory
- **Primary memory**
 - Information about which you are currently aware
 - Effortlessly available, but fleeting
 - Limited capacity
- **Secondary memory**
 - Memory of the past
 - Available with effort but permanent
 - Unlimited capacity



1842 - 1910

Updated version

Short-term memory

- Contents are highly accessible
- Brief and fragile
- Limited in capacity

Long-term memory

- Contents are less accessible
- Enduring
- Limitless capacity

- But... there's an even simpler version of temporary memory than James imagined

Sensory memory

- Our sensory systems convert sensory energy into neural representations
- Sensory systems have a brief “storage space” for these representations
 - Sensory memories are fleeting impressions of stimuli we *just* experienced



Sperling task (1960)

- Study of ***iconic*** memory (visual sensory memory)
- Briefly presented a grid of letters (50 ms)
- Participants immediately reported which letters they saw

+

O

P

T

J

Y

I

B

W

F

Q

U

Z

Sperling task (1960)

- Participants reported ~4-5 letters, but noted seeing more letters than they could report
- Sperling was interested in how much information was actually stored in iconic memory
- His hypothesis: iconic memory is **not** limited in *capacity*, but rather limited in *time*
 - Representations for the whole grid exist briefly
 - If we don't transform those representations to something more stable, we'll lose them

Sperling task (1960)

- Follow-up study:
 - **After** presenting the letter grid, a tone was played indicating which row of letters to report
 - If participants can report the cued row, there must be a brief representation of the *whole* grid

+

A	T	H	C
R	E	B	M
Z	O	W	L

Sperling task (1960)

- Results demonstrated that participants could, indeed, report letters from any row prompted by the tone
 - BUT, as soon as you start reporting from row X, you are simultaneously ‘forgetting’ information from rows Y and Z
- Sperling’s conclusions:
 - Sensory memory *is not* limited in capacity
 - When attending to sensory information, sensory memory is transformed into a more stable trace
 - However, our attention capacity is limited

Sperling task (1960)

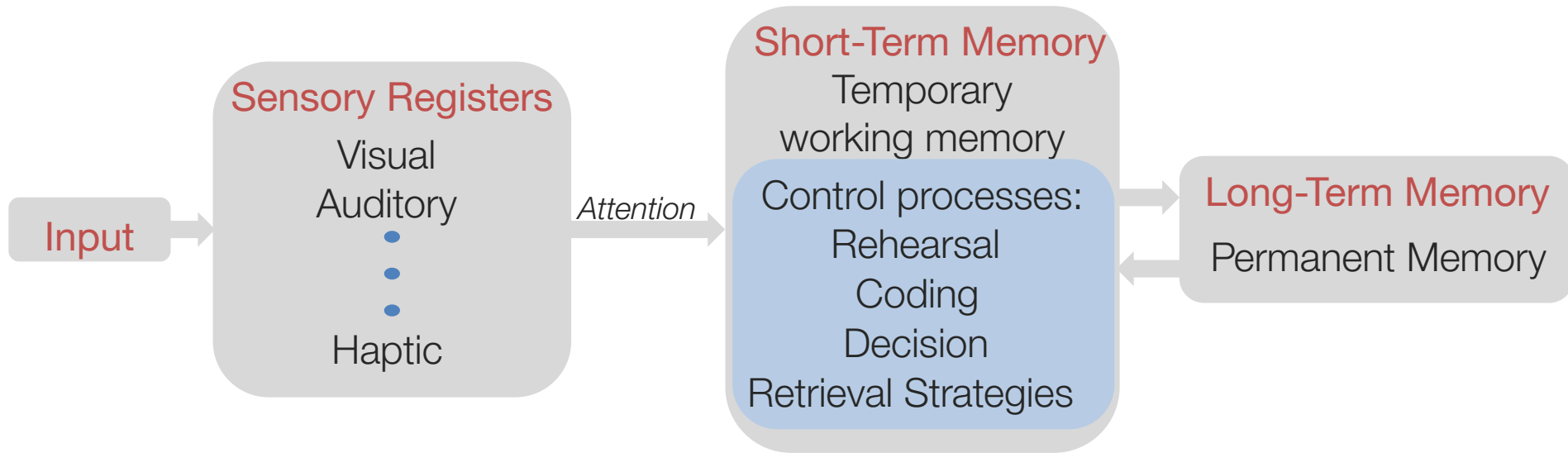


ANALOGY:

- Sensory memory is the 'unlimited' money in the booth
- Working memory is the amount of money your hands can grab and hold onto!

Atkinson-Shiffrin model (1968)

[aka the **Modal Model**]

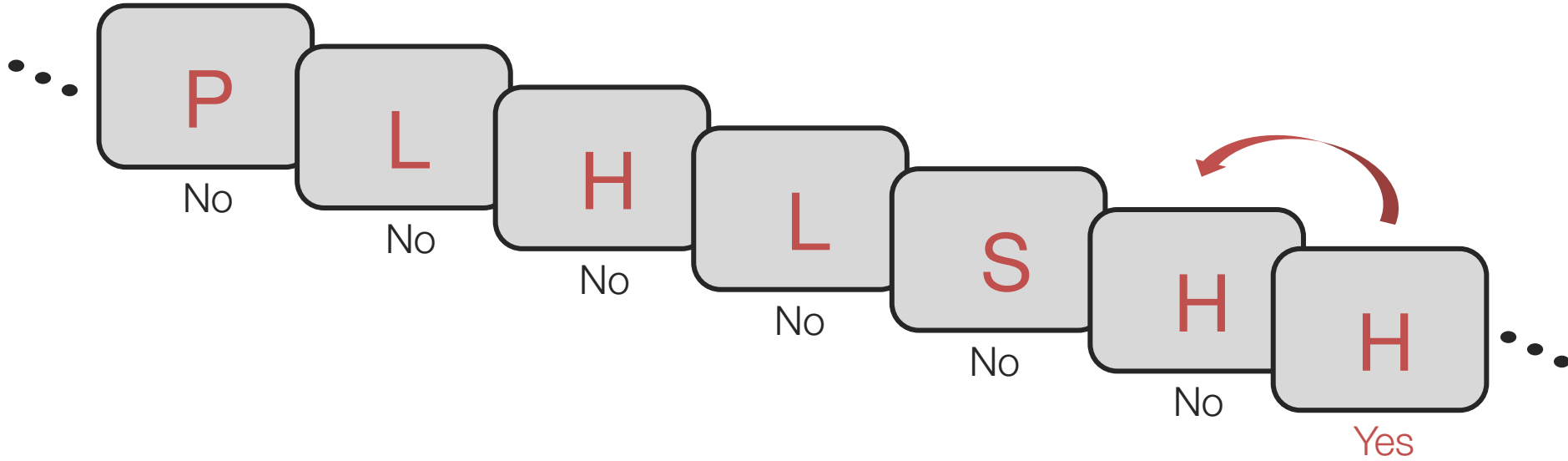


Working memory: a system for maintaining and manipulating active representations

- A shift from a *descriptive* term (STM) to a *functional* term (WM)

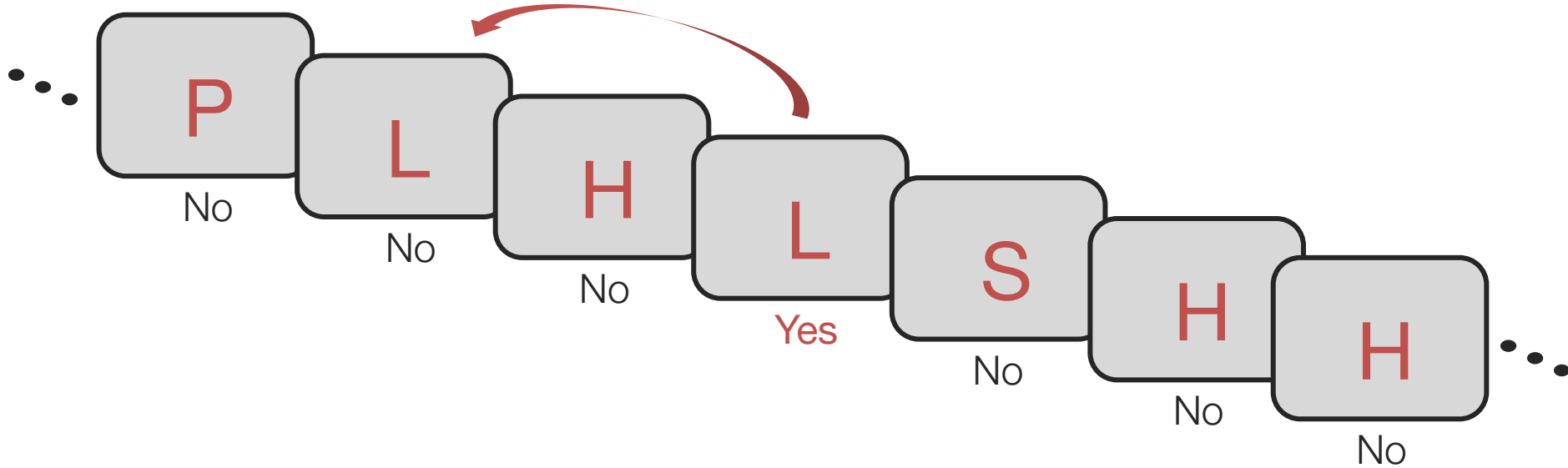
N-back task

- 1-back



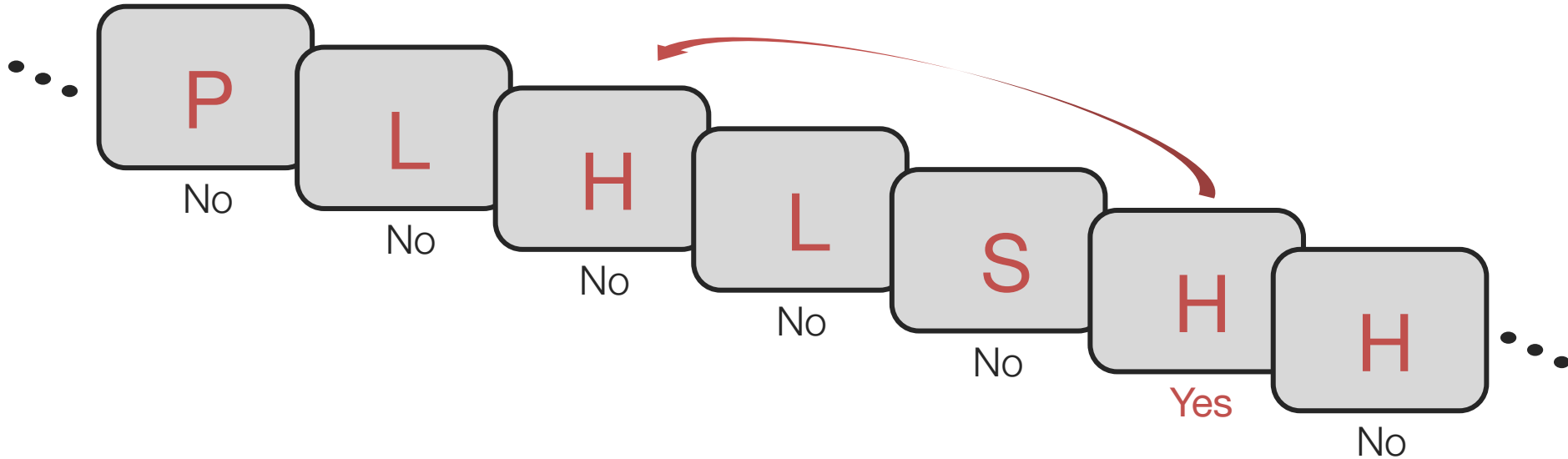
N-back task

- 2-back



N-back task

- 3-back



N-back task

Let's try a 2 back!
Clap when there is a 2-back match

N-back task

N-back task

O

N-back task

N-back task

L

N-back task

N-back task

B

N-back task

N-back task

H

N-back task

N-back task

B

N-back task

N-back task

A

N-back task

N-back task

Q

N-back task

N-back task

R

N-back task

N-back task

Q

N-back task

N-back task

F

N-back task

N-back task

F

N-back task

N-back task

N

N-back task

N-back task

F

N-back task

Working Memory Capacity

Outline

- What is WM?
- WM capacity
- Separate WM and LTM systems?

Measuring working memory capacity

- Capacity varies according to the working memory sub-system that is being measured
 - see article by Baddeley
- Generally, divided into phonological vs. visuo-spatial WM, each with different capacities
 - Many of the common tests of WM capacity measure capacity of phonological system (e.g. word lists)

Span tasks

- Working memory capacity is most frequently tested with *span* tasks (digits, letters, etc)
- Common approach:
 - Start with a single digit, and then increase the number of digits incrementally; how many digits can you remember?

Span tasks

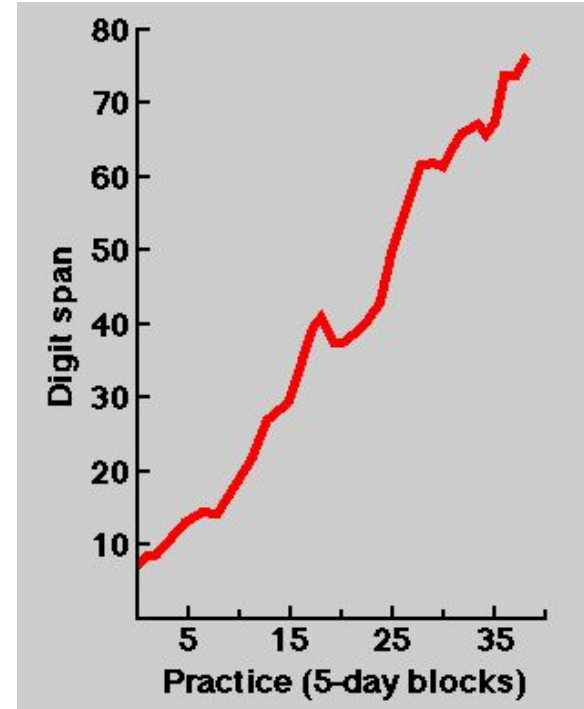
- George Miller (1956) suggested that there is a “Magic number 7, plus or minus 2”
- In other words, the vast majority of people have a working memory span between 5 and 9

Span tasks

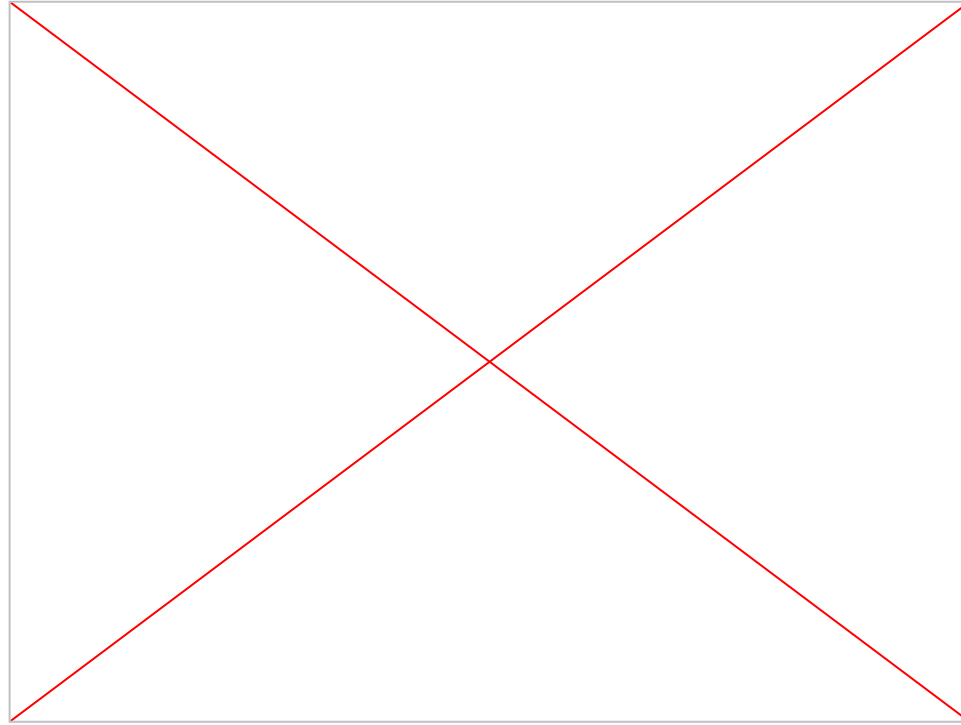
- With span tasks, we evaluate how many items are remembered... but what is an “item”?
- Try to remember the following string of letters:
 - C - A - T - F - L - Y - B - U - G
- What if the letters are reorganized as:
 - CAT-FLY-BUG
- Span increases by increasing the number of items in a “chunk”
 - Interplay between WM and LTM!

Ericsson and Chase (1982)

- Evaluated effects of chunking on a digit span task
 - Chunked numbers as running times
- With practice, “WM capacity” improved
- However, did not generalize to letter span



WM Capacity

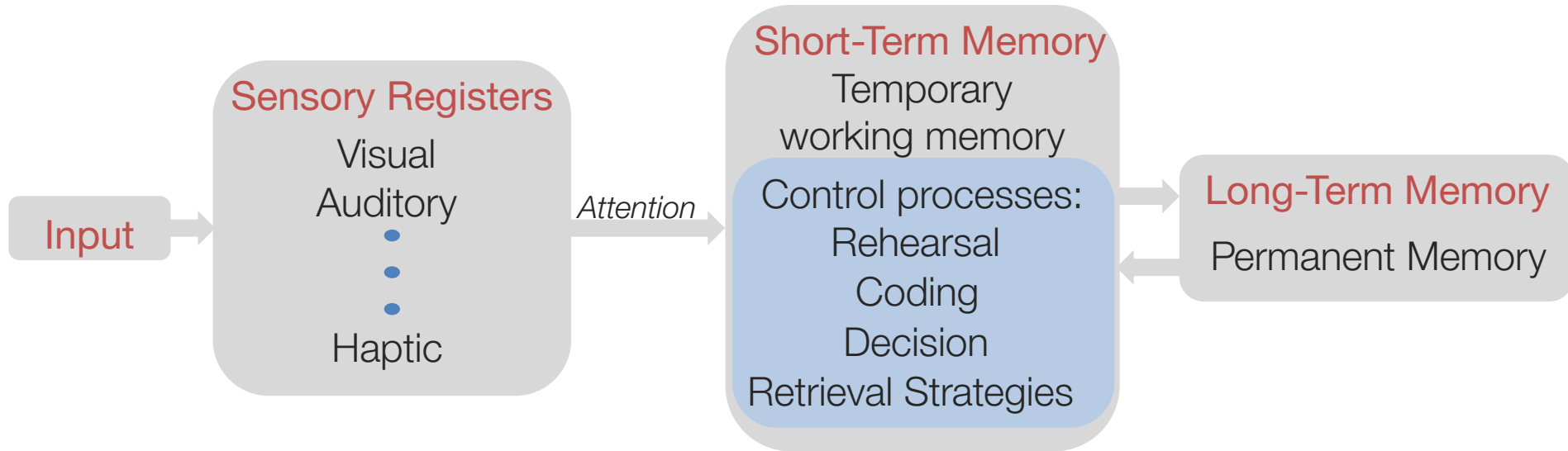


Working Memory vs. Long-Term Memory

Outline

- What is WM?
- WM capacity
- Separate WM and LTM systems?

Atkinson-Shiffrin model (1968)



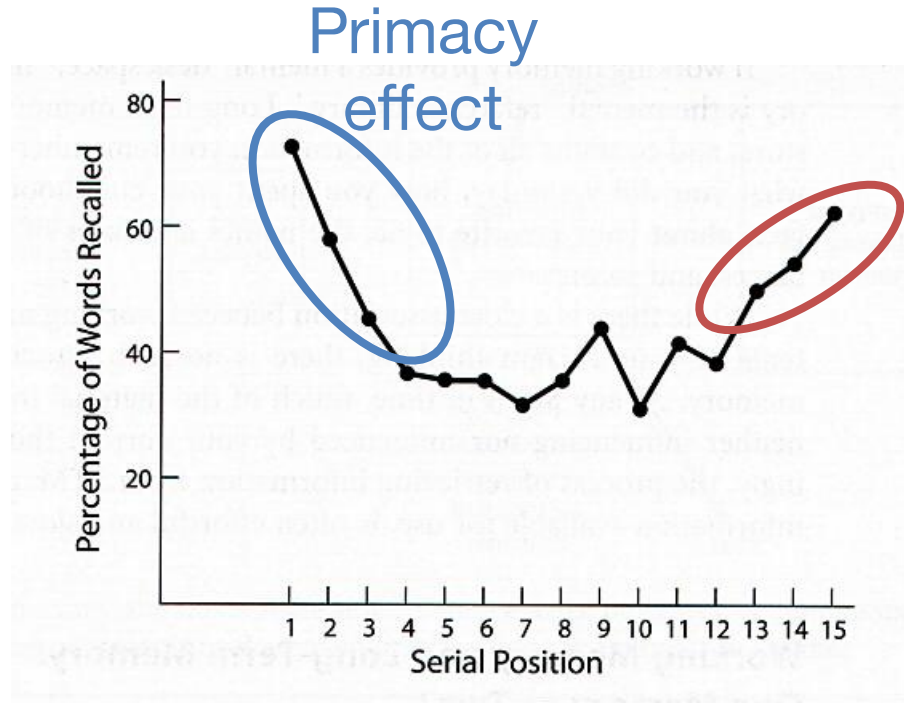
Multiple memory systems

What evidence do we have for separate STM and LTM systems?

- Much of this research has been conducted by looking at **serial position** and **free recall**
 - Serial position: ordering of items within a list
 - Free recall: read / listen to a series of words, then recall as many as you can

Free recall and serial position

1. Parade
2. Envelope
3. Balloon
4. Potato
5. Camera
6. Anchor
7. Umbrella
8. Taxi
9. Ladder
10. Glasses
11. Magnet
12. Broom
13. Sock
14. Dollar
15. Rabbit

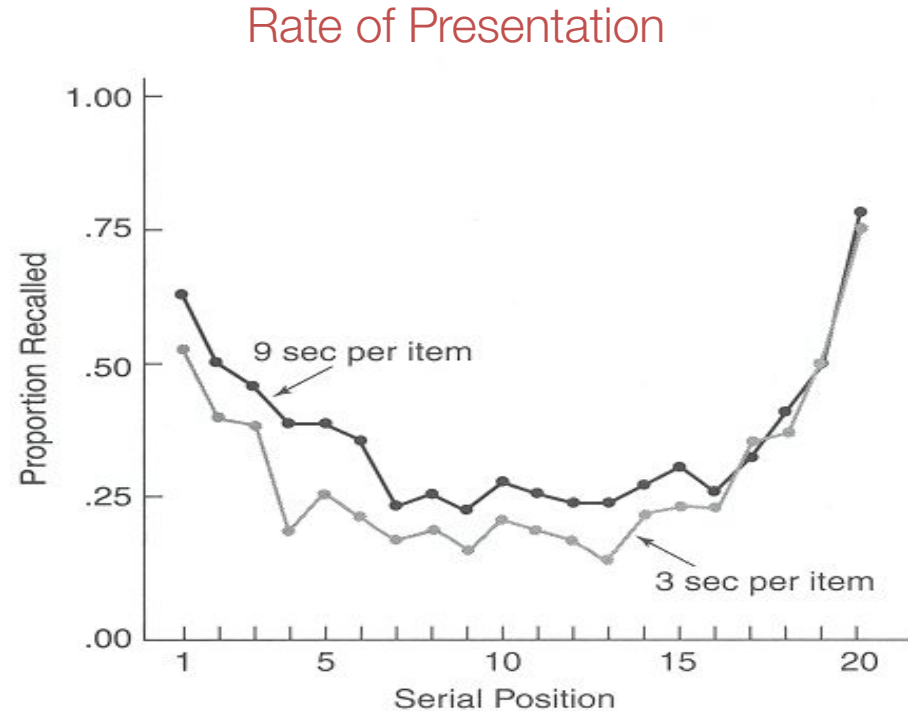


Recency effect

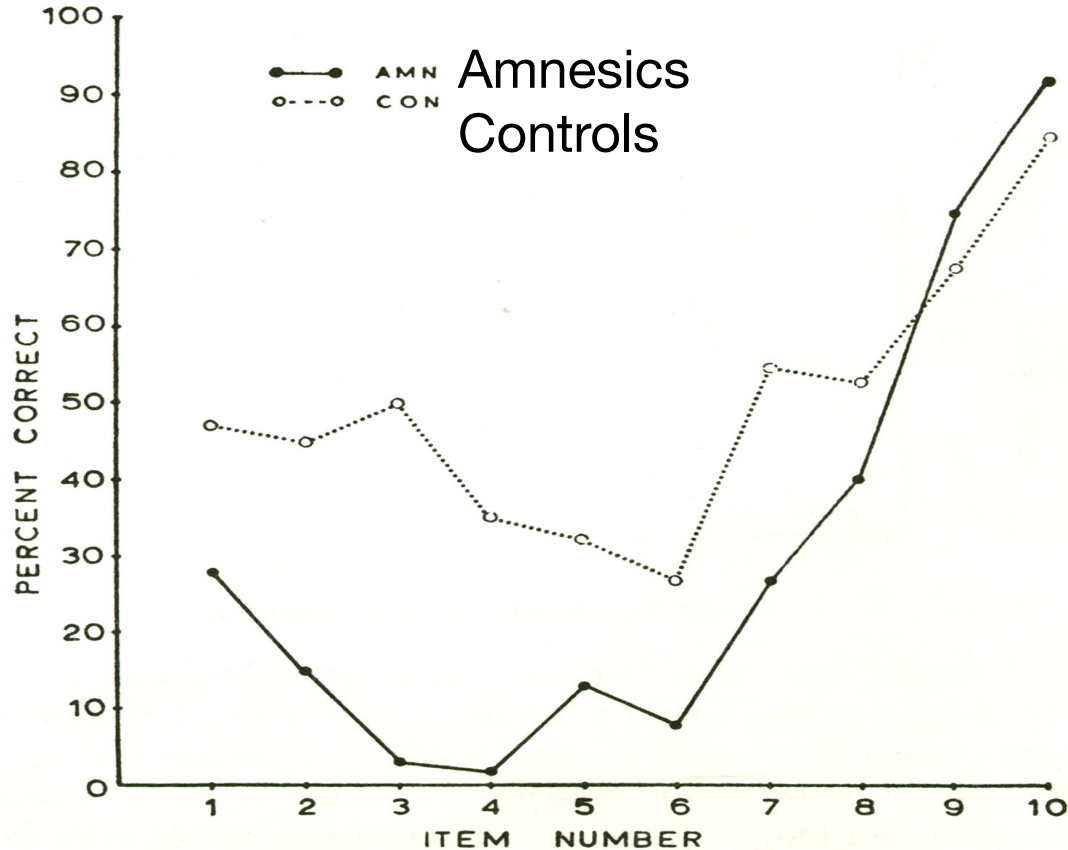
- Primacy effect reflects LTM; recency effect reflects WM

Factors affecting primacy

- **Rate** of presentation
- List **length**
- **Familiarity** of materials
- Medial temporal lobe **damage** (MTL)



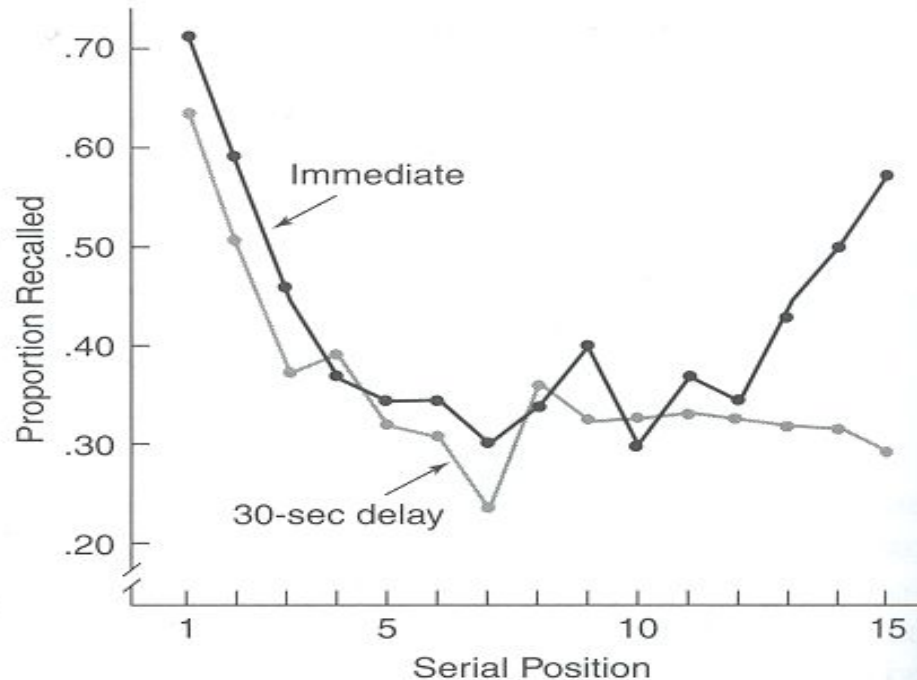
Medial temporal lobe damage



Factors affecting recency

- **Filled delay** (articulatory suppression) between study and test

Study-Test Delay



Multiple memory systems?

- Dissociations between STM and LTM using behavioral manipulations
 - Short presentation rate impairs primacy (LTM) but not recency (STM)
 - Filled delay impairs recency (STM) but not primacy (LTM)
- Dissociations between STM and LTM in patient populations
 - Patients with damage to medial temporal lobe show primacy (LTM) but not recency (STM) impairments
 - Patients with damage to frontal/parietal regions show recency (STM) but not primacy (LTM) impairments

Working Memory:

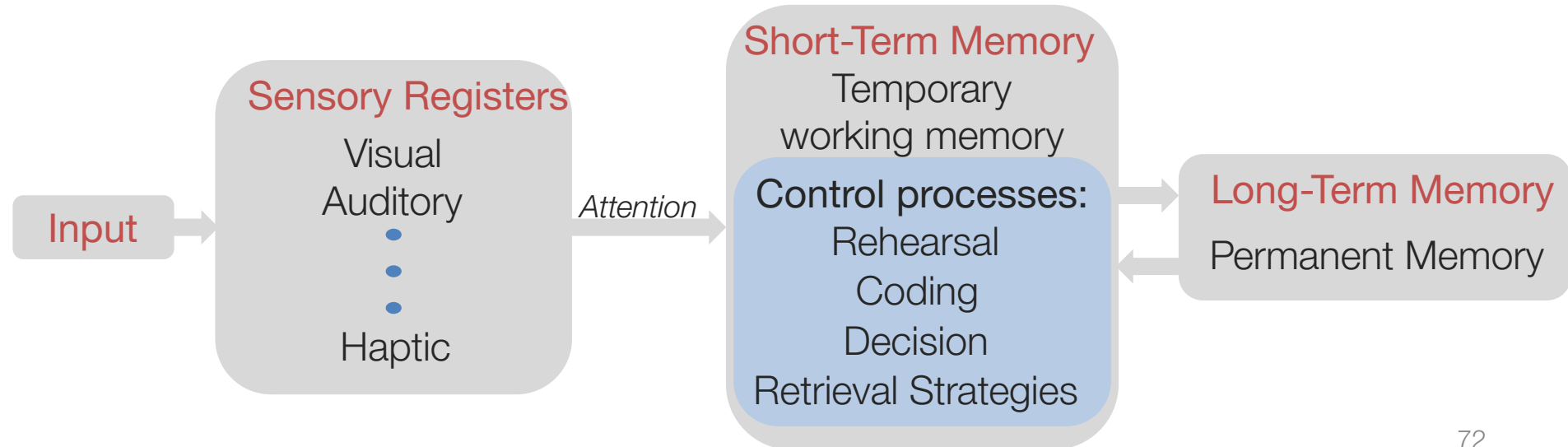
Phonological vs. Visuospatial systems

Outline

- Content effects in WM
- Tripartite model
 - Phonological and visuospatial buffers
 - Central executive and cognitive control
- Brain substrates

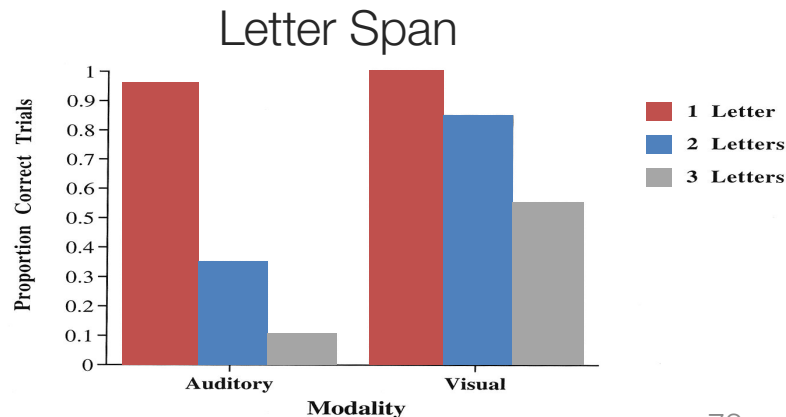
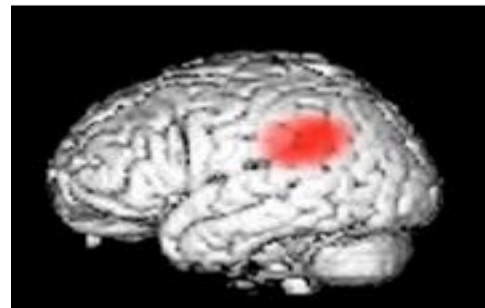
Is WM unitary?

- Atkinson-Shiffrin model
 - All sensory modalities and types of information pass through a common STM buffer
- But...
 - Behavioral, neuropsychological, and neuroimaging studies suggest that WM can be divided



Neuropsychological data

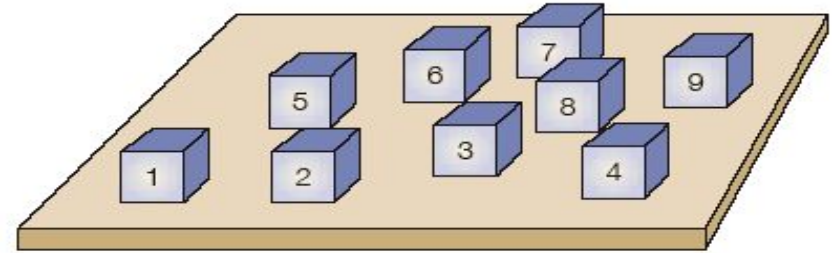
- Patient KF
 - Left hemisphere temporal/parietal lesion
 - Impaired auditory span relative to visual span



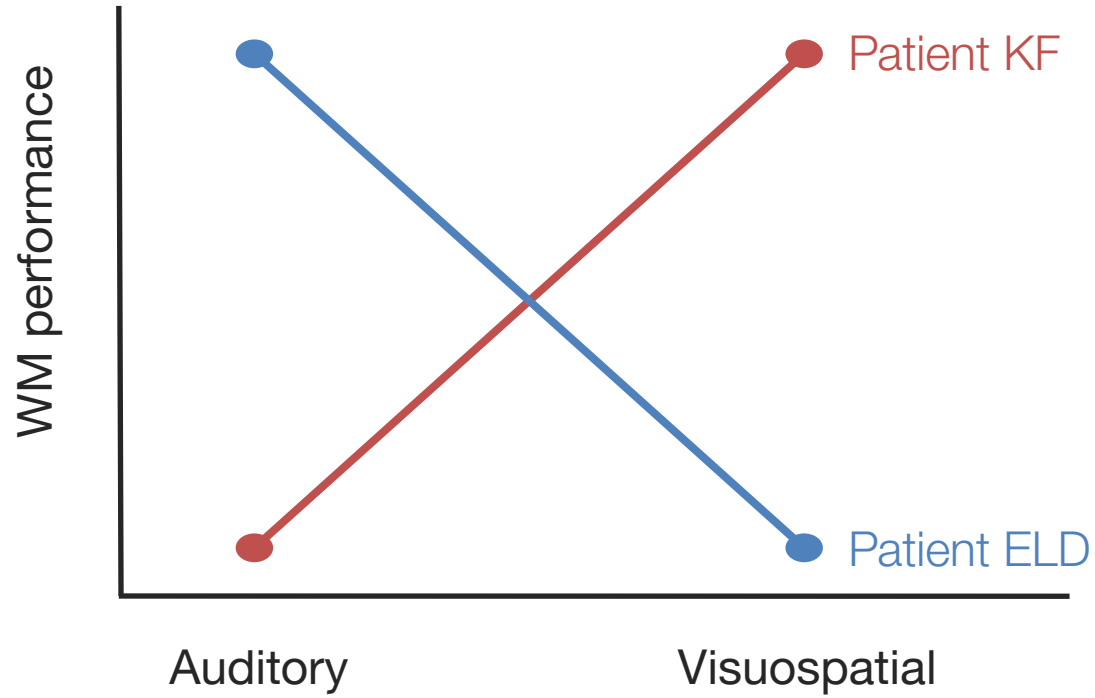
Neuropsychological data

- Patient ELD
 - Right hemisphere frontal/temporal lesion
 - Impaired visuospatial WM
 - Normal auditory WM

Corsi Block-Tapping Test



Double dissociation

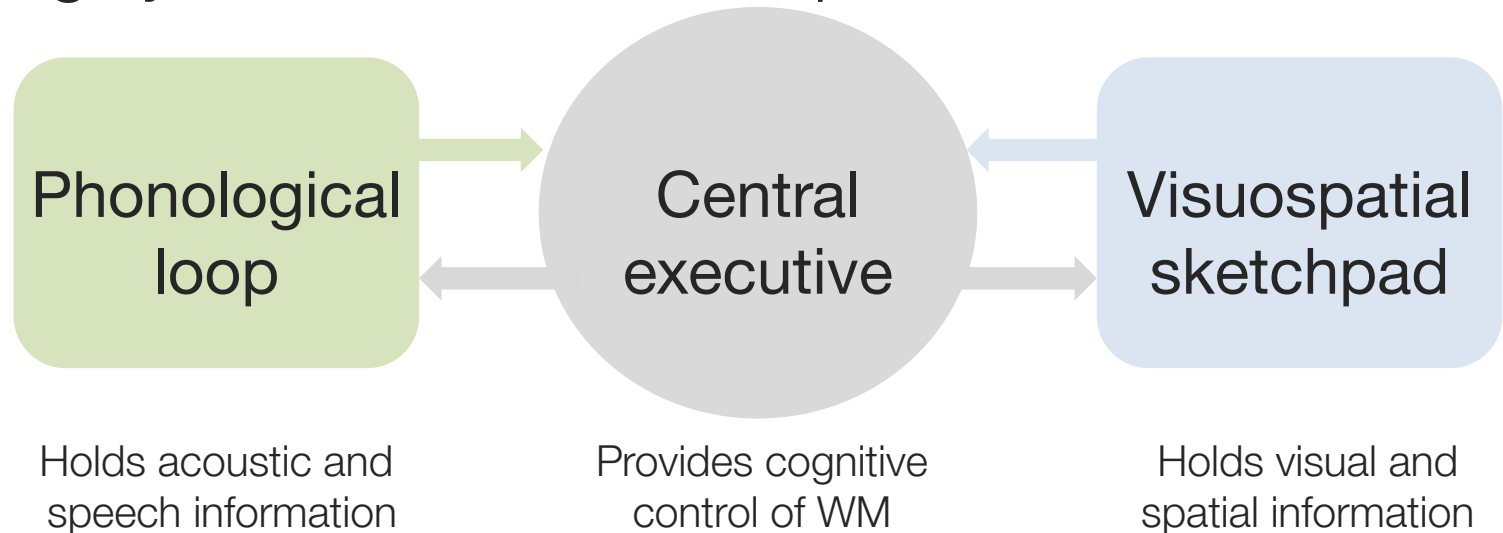


Outline

- Contents of working memory
- Tripartite model
 - Phonological and visuospatial buffers
 - Central executive and cognitive control
- Brain substrates

Tripartite model

- Baddeley and Hitch (1974)
- Highly influential multi-compartment model of WM



Tripartite model

Phonological WM

- Capacity of ~7, but varies according to material (e.g., *word length effect*)
- Directly related to sub-vocal articulation
- Predictive of language learning
- Impaired when material is phonologically similar

Visuo-spatial WM

- Capacity of 3 or 4 items
- May be related to spatial problem solving
- May be subdivided according to whether material is visual (e.g., a picture) or spatial (e.g., a sequence of locations)

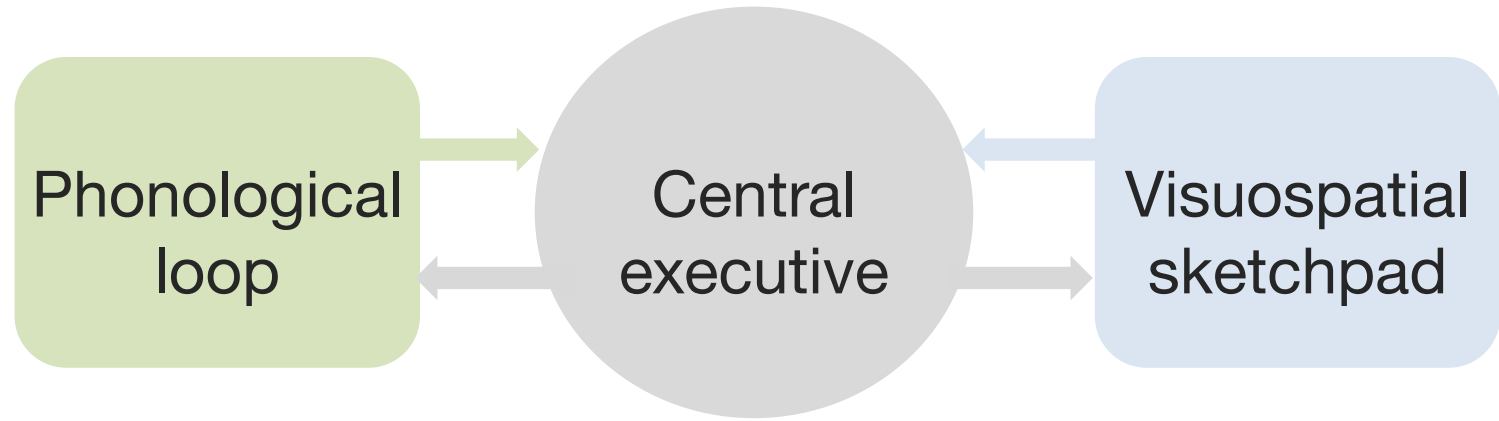
Working Memory:

Central Executive & Brain Substrates of WM

Outline

- Contents of working memory
- Tripartite model
 - Phonological and visuospatial buffers
 - Central executive and cognitive control
- Brain substrates

Central executive



- Exerts **control** over behavior
- Two main functions:
 - Decides which information is maintained
 - Manipulates the maintained information

Control processes

- Processes that flexibly shape and constrain our thoughts, memories, and actions to accomplish our goals
 - Inhibition
 - Selection
 - Shifting attention
 - Coordinating, updating, and monitoring working memory
 - Retrieving information from long-term memory

Stroop task

- Say aloud the **color of ink** in which each item is printed

more automatically retrieved information

Stroop task

- Say aloud the **color of ink** that each item is printed in
- XXXX much easier than Green
 - Control processes are required to override the automatic response (reading text)

Go / No-go task

- Whenever you see a letter, clap **as quickly** as you can... except if the letter is X

Go / No-go task

Go / No-go task

A

Go / No-go task

Go / No-go task

X

Go / No-go task

Go / No-go task

B

Go / No-go task

Go / No-go task

H

Go / No-go task

Go / No-go task

C

Go / No-go task

Go / No-go task

A

Go / No-go task

Go / No-go task

Q

Go / No-go task

Go / No-go task

R

Go / No-go task

Go / No-go task

K

Go / No-go task

Go / No-go task

U

Go / No-go task

Go / No-go task

O

Go / No-go task

Go / No-go task

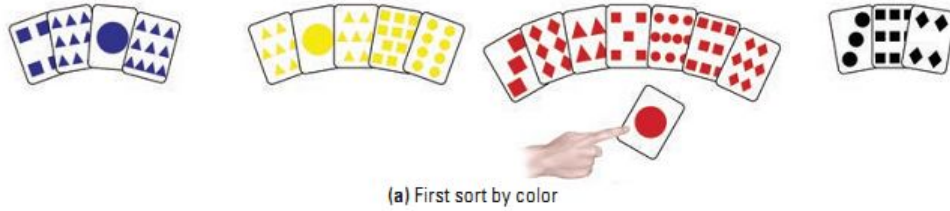
X

Go / No-go task

Go / No-go task

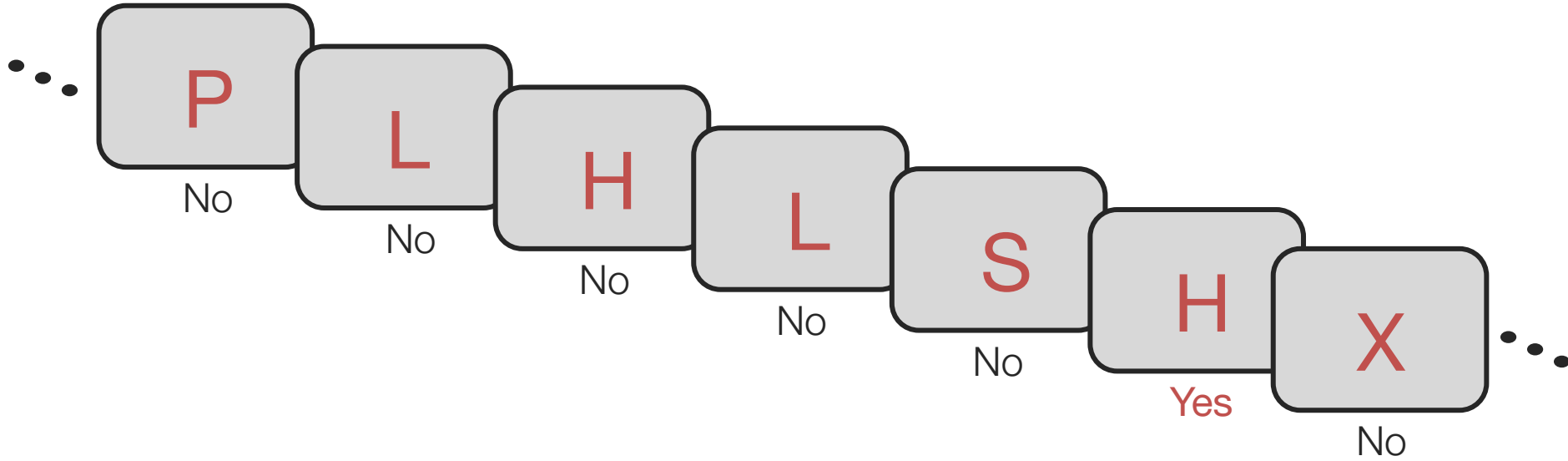
Given that X is *rare*, you have to inhibit the more dominant response

Wisconsin card sorting task

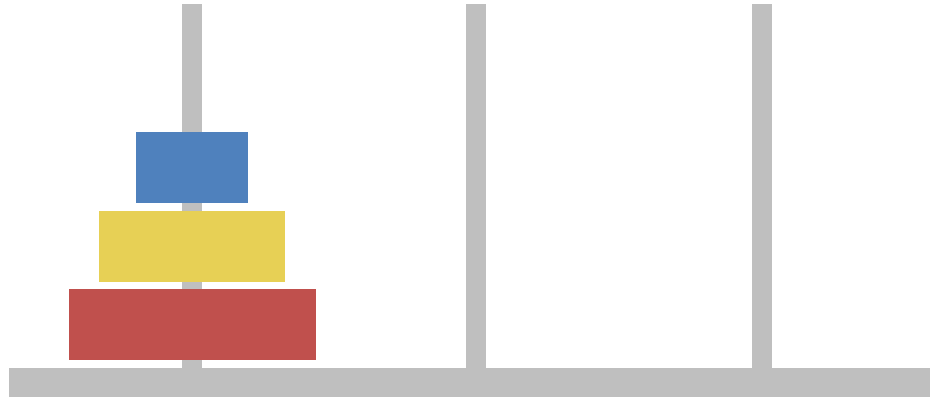


- Perseverative error: continuing to use old rule
- Successfully switching rules requires updating and inhibition

N-back task



Tower of Hanoi

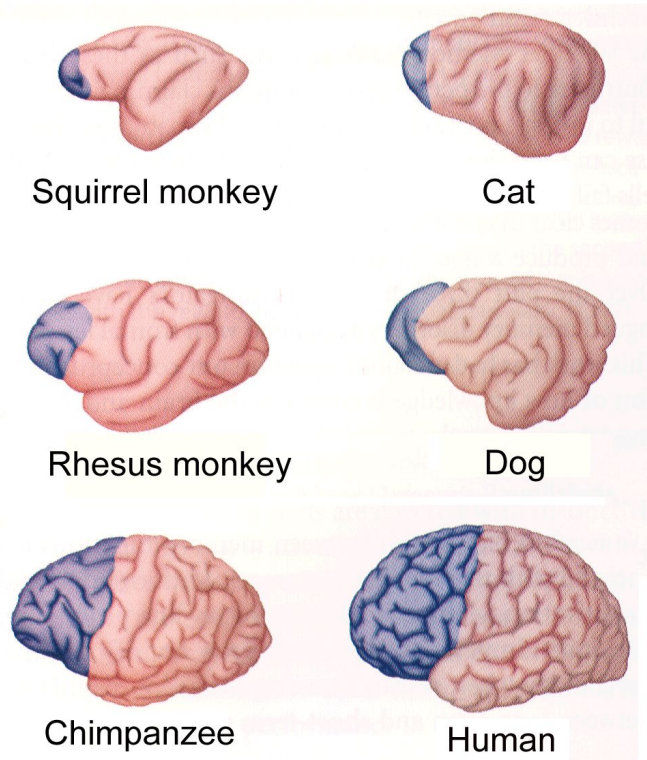


- Entire tower must be moved to the peg on the right
 - Move one disk at a time
 - Larger disks can't cover smaller disks
- Performance = number of moves
- Assesses control processes related to goals and planning

Outline

- Contents of working memory
- Tripartite model
 - Phonological and visuospatial buffers
 - Central executive and cognitive control
- Brain substrates

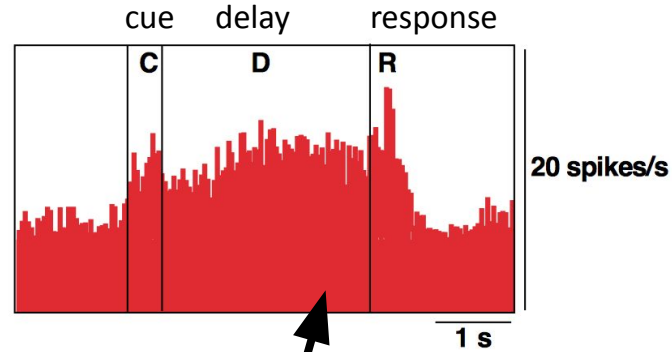
Prefrontal cortex (PFC)



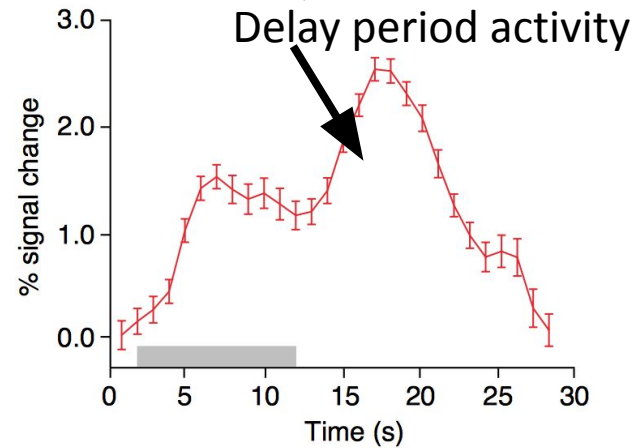
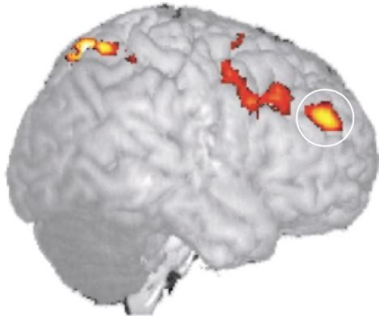
- Anterior section of the frontal lobe
- PFC is proportionally larger in the great apes than in other animals (~1/3 cortex)
- PFC plays a key role in WM and cognitive control

Prefrontal cortex (PFC)

(a) Macaque



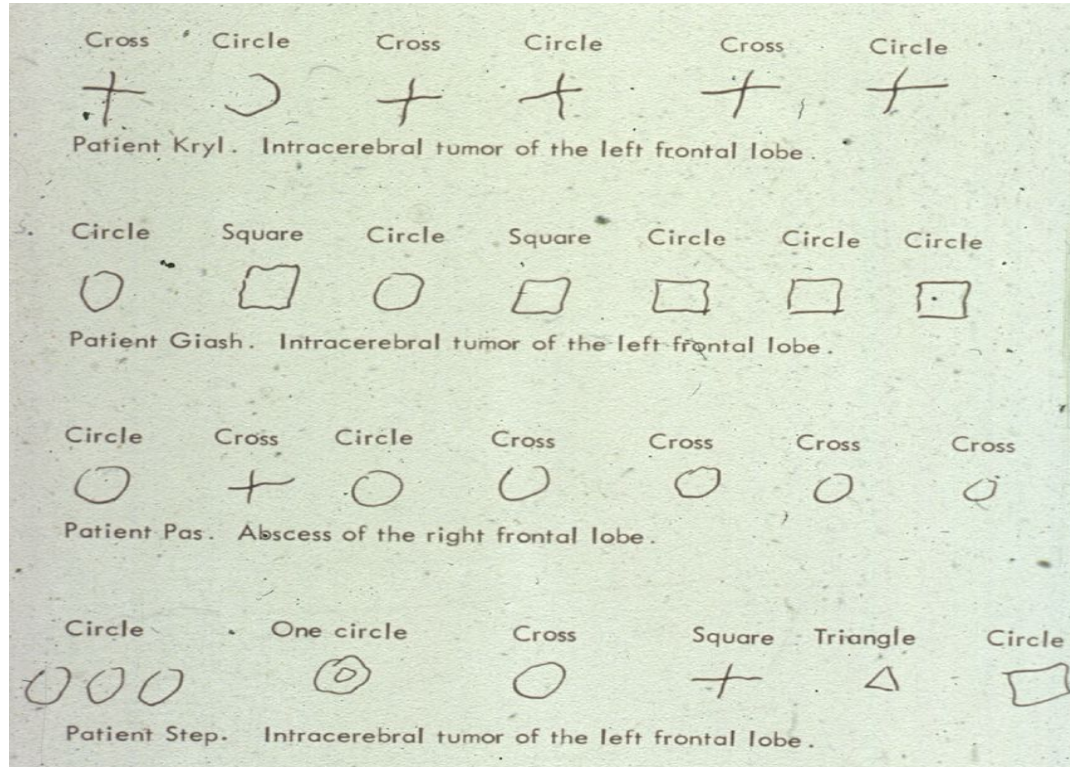
(b) Human



Damage to PFC

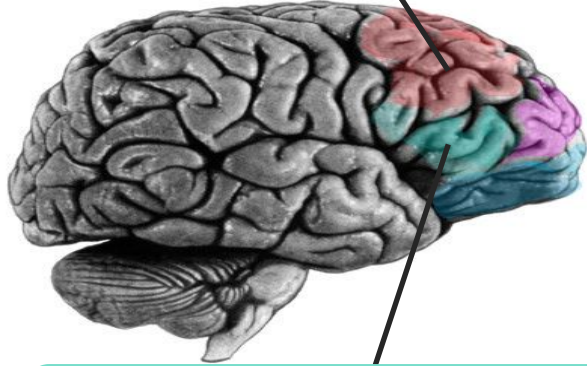
- PFC does more than just maintain information over a delay
- Patients with PFC lesions show deficits in WM and cognitive control
 - Increased distractibility
 - Perseveration
 - Difficulty inhibiting inappropriate responses
 - Inability to maintain information given interference
 - Difficulty strategizing and planning ahead
 - Difficulty initiating behaviors (inactivity)
- Collectively, these impairments are called **dysexecutive syndrome**

Perseveration



Subdivisions of PFC

Dorsolateral PFC



Ventrolateral PFC

- Two stage model of PFC (Petrides; Smith and Jonides)
 - Ventrolateral PFC (VLPFC)
 - First level of control
 - Interacts with posterior cortical regions to maintain information in WM
 - Dorsolateral PFC (DLPFC)
 - Second level of control
 - Monitors, updates, and manipulates information in WM

Evidence for two-stage model

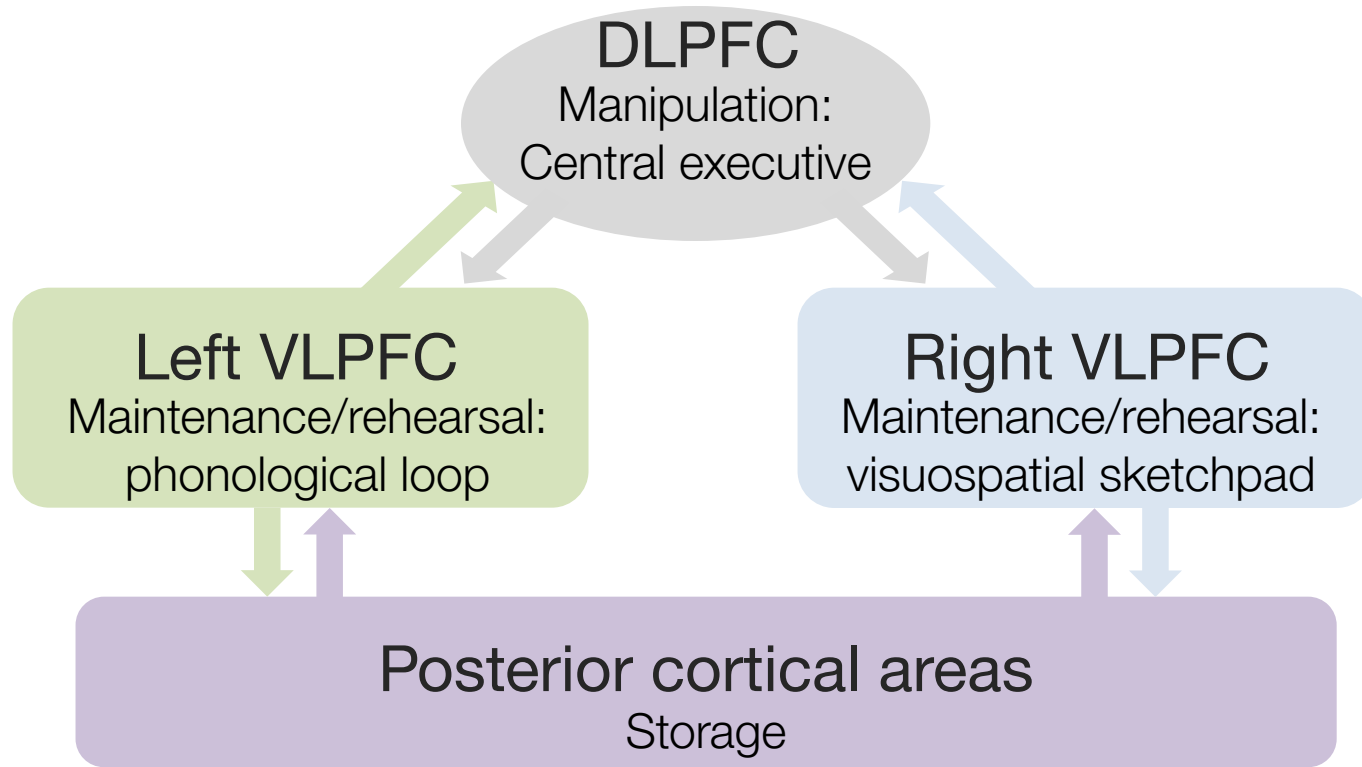
- Human neuroimaging data comparing PFC activity for two tasks:
 - Maintenance only
 - Maintenance, monitoring, and updating



Summary: Phonological and visuospatial buffers

- Behavioral data demonstrate little interference between phonological and visuospatial information during WM tasks
 - Suggests that different brain regions may support WM for different types of material
- Lesion data suggest hemispheric differences:
 - Left hemisphere lesions impair auditory WM more than visual WM
 - Right hemisphere lesions impair visuospatial WM more than auditory WM

Summary: Working memory in the brain



Next time...

Episodic memory

Reading on Canvas